IPGui Manual

Introduction:

When I began my career as an IT consultant in 1997, Windows 95 OSR2 was the standard operating system in the field. Back then, Microsoft included a handy tool called **WinIPcfg** — a graphical user interface (GUI) that made it easy to view and manage your computer's IP address and related network settings. For consultants and support staff, **WinIPcfg** was invaluable, especially when troubleshooting network problems or guiding users over the phone.

With the arrival of Windows XP in 2001, the classic 9x Windows line was retired—and with it, **WinIPcfg** disappeared. Its replacement, ipconfig, is a powerful tool, but it's command-line only. Many users, especially those less comfortable with technical details, found this a step backward in usability.

That's why I created **IPGu**i: to bring back the simplicity and clarity of **WinIPcfg**, but for modern Windows systems. My goal was to provide a straightforward, user-friendly tool for anyone dealing with IP address issues—whether you're an end user, a helpdesk technician, or an IT consultant working remotely with clients.

But **IPGui** is more than just a replacement for **WinIPcfg**. Over time, I've enhanced it with a wide range of advanced network tools, making it a powerful utility for both basic troubleshooting and in-depth network analysis. With features like live network usage monitoring (including a unique trip counter), WiFi scanning, ARP table management, port and IP scanning, and much more, **IPGui** is designed to be your all-in-one network toolbox—accessible to everyone, yet powerful enough for professionals.

And best of all, **IPGui is completely free**—both as in "free beer" and "free speech."
There are no hidden installs, no ads, and no unwanted extras—just a helpful program you can use however you like.
If you decide to create your own version, please remember to respect the GPL V2.

Enjoy using **IPGui**, and happy networking!

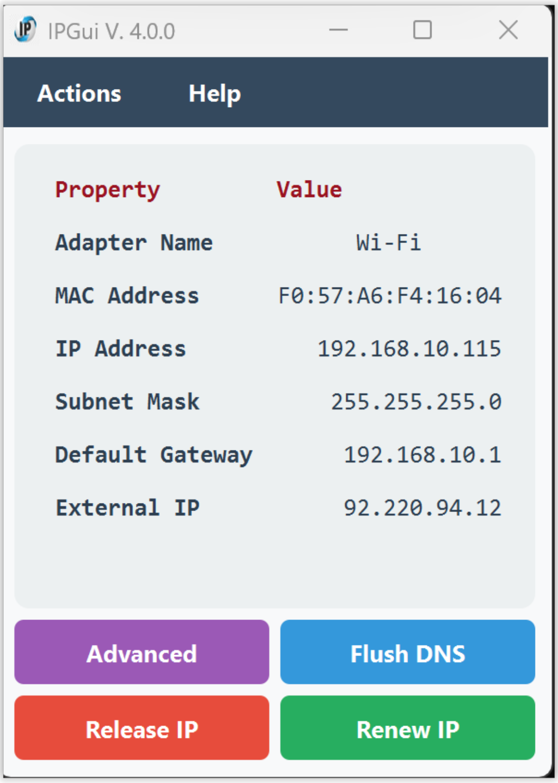
Note: This is an electronic document. The content listing on next page is alfabetized. The Entries in the manual, however was added as I made them. That makes this document less suited for print out, since the entries is not in the same order as the Content List.

IPGui Manual

Content:

The first Window:

Let's look at the first window when you start the program.



There is a menu on the window top, but we will look on that later. Right now this page is what we are consentrating of.

First you will see the «Adapter name». That is the name of your active network card, the card your machine speaks with the network through.

Then you will see the card's MAC Address. That is a set of hexadecimal numbers divided bay a colon. This address is unique to your network card.

Below that, you find the «IP address» of your computer This is given you via DHCP (Dynamic Host Configurating Protocol), and is the address of your machine in the network.

Next, you find the «Subnet Mask». Usually in home networks it will be 255.255.255.0 since you have only one subnet. In companies (and some geeks' home), there are several sub networks that is controlled by one router. Then this comes into play to separate the locel networks.

After the Subnet Mask, you find the «Default Gateway». In a normal home, that's always your router. It is your gateway out and in of your network.

At last in this listing you find your «External IP address». That is the address that the world talk to your router through. And then the router sends the information either from you to the internet, or from the internet to you.

Below the listing there are four buttons. These buttons are the ones the user needs most when it comes to DHCP.

→ **Advanced**
This will give the same output as «ipconfig /all» at the command line. An ordinary user will not need that, but it is known for any IT consultant and his troubleshooting of the network.

- **Flush DNS**
  When you flush the DNS, you are clearing the cache that your computer uses to store information about domain names and their corresponding IP addresses. The DNS cache will the rebuild itself. So you will usually don’t notice it. It might solve problems with your DNS cache though.

- **Release IP**
  What it does:

  Releases Your IP Address: When you run this command, your computer tells the *DHCP server* (the device that assigns IP addresses) to give up the current IP address it has. Think of it like returning a library book you no longer need.

  Makes IP Address Available: By releasing the IP address, it becomes available for other devices on the network to use. This is useful if you want to change your network settings or if there’s a problem with your current IP address.

  Prepares for a New IP Address: After you release the IP address, you typically follow it up with the *Renew IP* button This command asks the DHCP server for a new IP address, like getting a new library book.

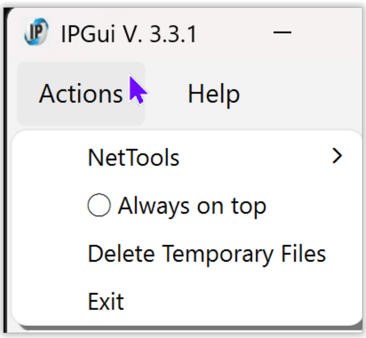
- **Renew IP**
  Requests a New IP Address: When you run this command, your computer asks the *DHCP server* (the device that assigns IP addresses) for a new IP address. Think of it like asking for a new library book after you’ve returned the old one.

Re-establishes Connection: This command is typically used after you’ve released your old IP address with the It helps your computer reconnect to the network with a fresh IP address.

Fixes Network Issues: If you were experiencing connectivity problems, renewing your IP address can often resolve these issues by providing a new address that might work better.

As a normal user, this window is all you need. But I have included a lot of other functions in the **«Action»** menu to work with and diagnose your network.

This is the «Actions» menu:



We do not start with the «NetTools» submenu before we have looked at the two others, so we start at the bottom of the menu.

Exit, simply exits the program so you can get on with whatever you were doing.

Always on top

The Always on top option keeps the **IPGui** main window and all NetTools dialogs (such as Ping, IP Scanner, NS Lookup, etc.) visible above all other windows on your desktop.

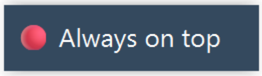
Advantages:

You can monitor your network info or use network tools while working in other programs, without the **IPGui** windows getting hidden behind other windows.

All NetTools dialogs (Ping, IP Scanner, NS Lookup, Traceroute, Port Scan, etc.) will also stay on top, making it easy to compare results or copy information while multitasking.

How to use:

- Click Actions → Always on top in the menu to turn this feature on or off.
- When enabled, a red icon (image at the right) appears next to the menu item; when disabled, a gray circle (◯) is shown.



- To turn it off, just select the menu item again.

Best use:

Enable Always on top when you want to keep **IPGui** and its tools visible while switching between other applications, such as your web browser, text editor, or remote desktop.

This is especially helpful when troubleshooting, copying results, or monitoring network changes in real time.

Tip:

If you find the windows are getting in the way, simply turn off «Always on top» to return to normal window behaviour.

Delete Temporary Files.

Purpose:

This function allows you to quickly remove all temporary files created by **IPGui** in the folder:
C:\Users\Public\AppData\Local\IPGui

This is especially useful when running **IPGui** as a portable application, as it cleans up any leftover data, logs, or backups that may have been created during use.

How to Use:

- Open the Actions menu in the main window.
- Click "Delete Temporary Files".
- A confirmation dialog will appear:
  «Are you sure you want to delete all temporary files for this application? This will remove: C:\Users\Public\AppData\Local\IPGui»
- Click "Yes" to proceed, or "Cancel" to abort.
- If the folder exists and is deleted successfully, you will see:
  «Temporary files deleted.»
- If the folder does not exist, you will see:
  «No temporary files found.»
- If deletion fails (e.g., files are in use), you will see:
  «Failed to delete some or all temporary files.»

Notes:

No administrator rights are required.
The folder is in a location accessible to all users, and deletion is performed with normal user permissions.

Safe to use:
If the folder is missing or already empty, nothing will be deleted and you will be notified.
Automatic recreation:
The folder and any needed files will be recreated automatically by IPGui the next time they are needed (for example, when you open the Hosts File Editor or download the manual).

Portable use:
This feature is especially helpful when using IPGui as a portable app, to ensure no traces are left on the system after use.

Tip:
If you want to fully remove all traces of IPGui after using the portable version, use this function before closing the application.

The NetTools sub-menu:

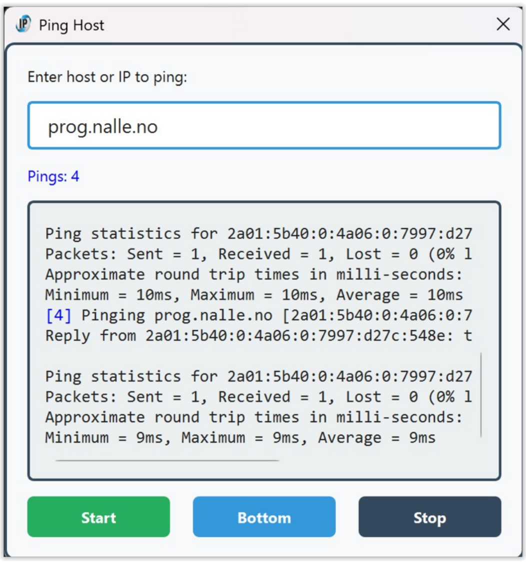
When troubleshooting network problems one need a lot of network tools to diagnose and solve the problem at hand.

I have tried to include as many of those as I can, to simplify the search for an odd network error.

Not to disturb the simplicity of the front window, I have «hidden» it in the sub menu.

Here are the tools:

Ping



Ping lets you check if a computer or device on the internet or your local network is reachable.

Just enter a hostname (like www.google.com) or an IP address (like 8.8.8.8) and click Start.

The tool will send a series of test messages ("pings") and show you if the device responds, how long it takes, and if any packets are lost.

Typical uses:

- See if a website or server is online.
- Test your connection to your router or another device on your network.
- Diagnose network problems.

How to use:

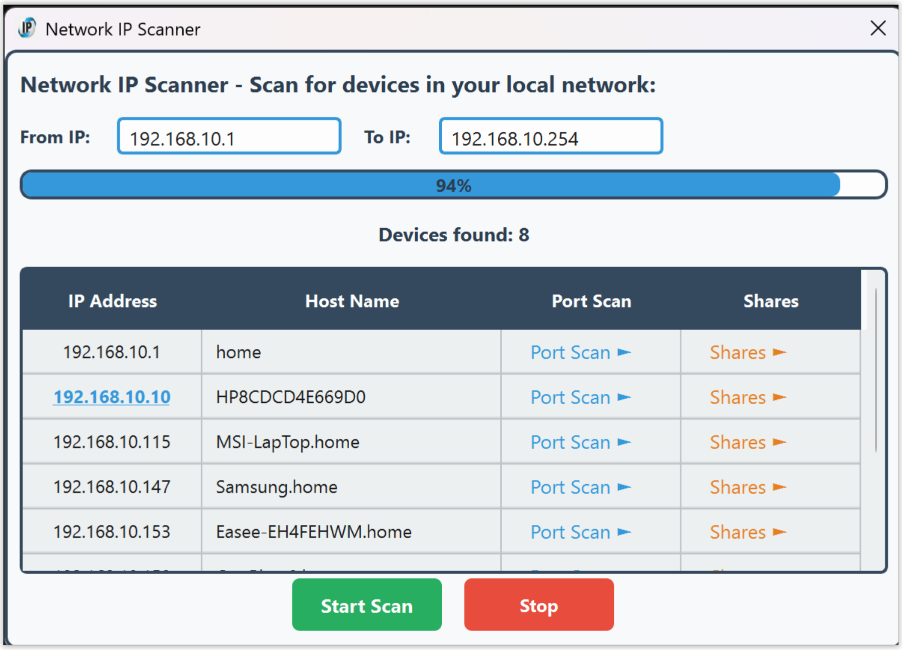
- Go to Actions → NetTools → Ping…
- Enter the address you want to test.
- Click Start to begin pinging.
- Click Stop to end the test, or Close to exit.

Tip:

If you see replies with times (in ms), the device is reachable.

If you see "Request timed out" or no replies, the device may be offline or blocked by a firewall.

IP scanner



The IP Scanner helps you quickly find all devices connected to your local network.
It scans a range of IP addresses and shows which ones respond, along with their names (if available).

If a device has a web interface, you can click its IP address to open it in your browser.

What else can you do with the IP Scanner?

For each device found, you get two extra buttons:

Port Scan

Click this to check which common services (like web, file sharing, or remote access) are open on the device.

This helps you see if a device is running a web server, printer service, or other network features.

Example: You can check if your printer is sharing files, or if your router has a web admin page.

Shares

Click this to see if the device is sharing any folders (Windows file shares).

You'll get a list of shared folders, and you can copy the network path to access them directly.

Example: Find shared folders on another computer or a network drive, and open them easily.

Why are these features useful?

Easier troubleshooting:
If you can't find a device or access its files, you can quickly check if it's online, what it's sharing, and what services are available.

Better overview:
See all devices, their names, and what they offer on your network in one place.

Quick access:
Open web interfaces or shared folders with a single click, without guessing addresses or paths.

Typical uses

- See all computers, phones, printers, and smart devices on your network.
- Find the IP address of a device (like a router, camera, or smart TV).
- Check which devices are currently online.
- See what services (web, file sharing, etc.) are available on each device.
- Quickly access shared folders on other computers.

How to use

Go to Actions → NetTools → IP Scanner…

The tool will suggest a range of IP addresses based on your network.

You can adjust the range if needed.

Click «Scan» to start scanning.

Devices found will appear in a list.

If a device has a web interface, its IP will be a clickable link.

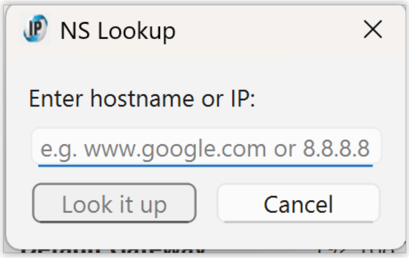
For each device, use:

- Port Scan to see what services are open.
- Shares to see and access shared folders.
- Click Stop to end the scan early, or Close to exit.

Tip:

If you don't see a device, make sure it's powered on and connected to the same network.

NS Lookup



NS Lookup lets you find the IP address for a website, or find the website name for an IP address.

Just enter a hostname (like www.google.com) or an IP address (like 8.8.8.8) and click Look it up.

The tool will show you the DNS server used, the official name, and all IP addresses found for that host.

- **Typical uses:**
  Find out the IP address of a website.
- Check which website a certain IP address belongs to.
- Troubleshoot DNS problems.

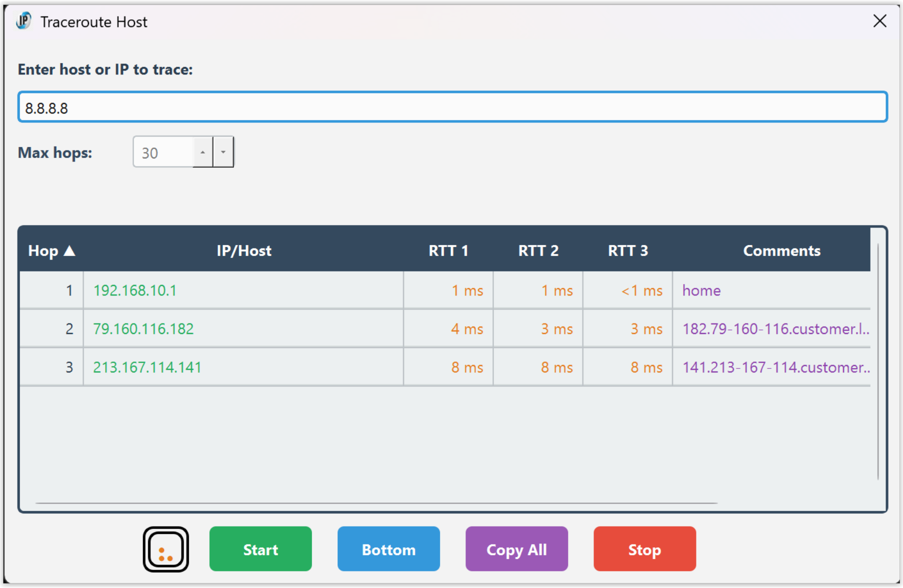
How to use:

- Go to Actions → NetTools → NS Lookup…
- Enter the website name or IP address you want to look up.
- Click Look it up.
- The result will show the DNS server, the name, and all addresses found.

Tip:

If you get no result, check that you typed the address correctly or try another website.

Traceroute



Traceroute shows you the path your network traffic takes to reach a website or IP address. It lists all the routers (called "hops") your data passes through on its way to the destination, displaying three timing measurements and detailed information for each step.

Advantages:

Visual network mapping:
See the complete route with color-coded entries for easy identification

Detailed timing analysis:
Three RTT (Round Trip Time) measurements per hop for accuracy

Professional presentation:
Clean, sortable table with proper hostname and IP separation

Troubleshooting power:
Identify exactly where delays or problems occur in the network path

How to use:

- Go to Actions → NetTools → Traceroute...
- Enter the website name or IP address you want to trace
- (Optional) Set the maximum number of hops (steps) to check (default: 30)

- Click Start to begin the trace
- Watch the animated progress indicator and real-time results

View detailed information in the color-coded table:

Green rows: Private network hops (local network)
Blue rows: Public internet hops
Red rows: Timeouts or unreachable hops

- Use **Bottom** to scroll to the latest results
- Use **Copy All** to export results in CSV format for analysis
- Click **Stop** to end early, or **Close** to exit

Best use:

Use Traceroute when a website is slow or unreachable. If the trace stops before reaching the destination, the problem is likely at the last successful hop shown. The color coding helps you quickly identify whether issues are in your local network (green) or on the internet (blue/red).

Pro Tips:

The Copy All button exports results in CSV format - perfect for sharing with IT support or further analysis

Sort results by clicking column headers (Hop, IP/Host, RTT values, Comments)

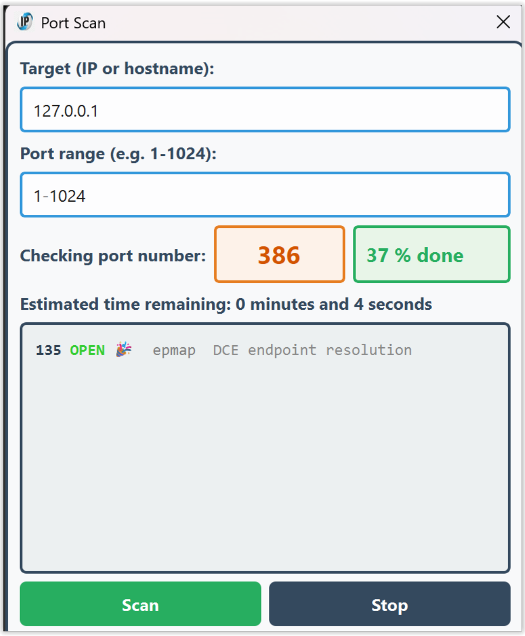
Three RTT measurements provide more reliable timing data than single measurements

Hostnames in the Comments column help identify ISPs and network providers along the route

Port scan

The Port Scan tool checks which network ports are open on a device (such as a computer, server, or router).

Open ports are like doors into a device—some are needed for services (like web or file sharing), but others may be open by mistake or for unwanted programs.



Advantages and uses:

Security:
Find open ports that could be used by hackers or malware to access your device. You can then close or secure any ports you don't need.

Troubleshooting:
Check if a service (like a web server or game) is reachable from your network.

Network management:
See which services are running on your computers or devices.

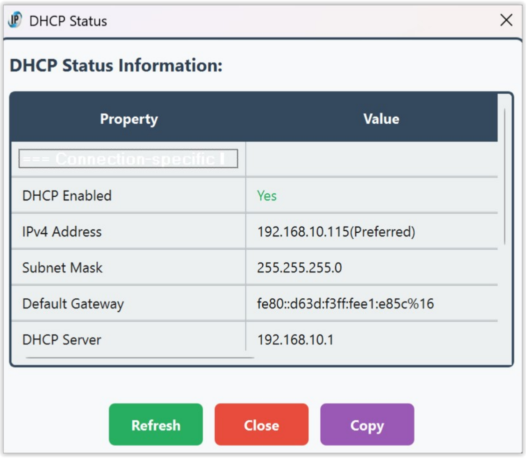
How to use:

- Go to Actions → NetTools → Port Scan…
- Enter the IP address or hostname of the device you want to scan.
- Enter the range of ports to check (for example, 1-1024).
- Click Scan to start. Open ports will be listed, along with their common service names and descriptions.
- Click Stop to end the scan early, or Close to exit.

Tip:

If you find open ports you don't recognise, consider closing them in your device's firewall or settings for better security.

DHCP status



The DHCP Status tool shows you details about your current network connection and how your computer gets its IP address.

What it does:

- Tells you if your computer is using DHCP (automatic IP assignment).
- Shows which DHCP server gave you your IP address.
- Displays when your current IP lease started and when it will expire.
- Shows your network adapter's name and DHCP client ID.

Why use it?

- To check if your computer is set up to get its IP address automatically.
- To see which server is managing your network connection.
- To troubleshoot network problems, especially if you're not getting an IP address or internet access.

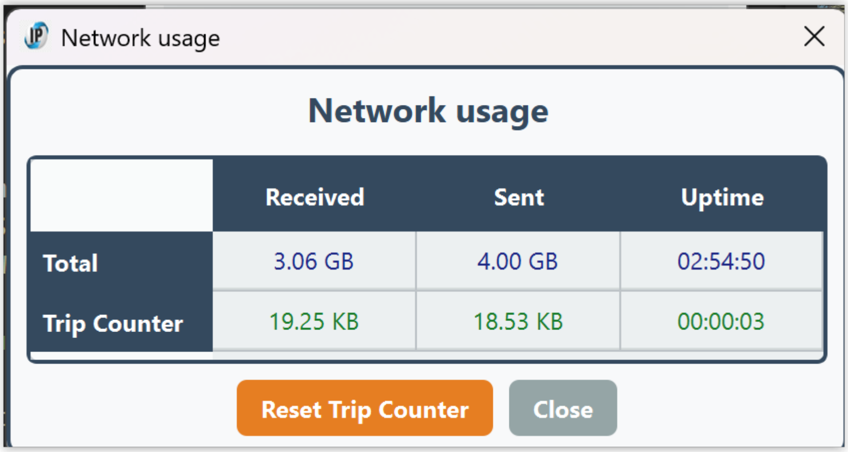
How to use:

- Go to Actions → NetTools → DHCP Status…
- The tool will show a table with your network adapter, DHCP server, lease times, and other details.
- Click Close to exit.

Tip:

If you see "No active DHCP-enabled interface found," your computer might be using a static (manual) IP address or not connected to a network.

Network Usage



What does this window do?

This window shows you how much data your computer has sent and received over the network.

It also shows how long your network has been running.

What do the numbers mean?

Total:
Shows how much data you've sent and received since you started your computer or connected to the network.
Also shows how long your network has been up (Uptime).

Trip Counter:
Lets you measure data for a specific period (like a stopwatch for your network).
Shows how much data you've sent and received since you last pressed "Reset Trip Counter".
Shows how long the trip counter has been running.

How do I use the Trip Counter?
Click "Reset Trip Counter"

A box will pop up asking how long you want to measure for.

Enter the time as hh:mm:ss (for example, 1:00:00 for 1 hour, 0:30:00 for 30 minutes, 0:00:10 for 10 seconds, or just 0 for unlimited time).
If you type something wrong, you'll get a warning and can try again.

Watch the Trip Counter row:
It will start counting from zero.

The numbers show how much data you've used during your chosen time.

When the time is up the numbers on the Trip Counter line will turn red.

This means the timer has stopped and the numbers are now "frozen".

You can still see how much data you used in that period.

To start again:
Just click «Reset Trip Counter» and set a new time.

Why is the Trip Counter useful?

Want to know how much data YouTube uses in an hour?

Set the trip timer for 1:00:00, watch your video, and check the red numbers when it's done!

After one hour the counter will stop and the numbers turn red. It will stay that way until you press «Reset Trip Counter».

Want to measure your data for a download, a game, or a Zoom call?

Use the trip counter for any period you want.

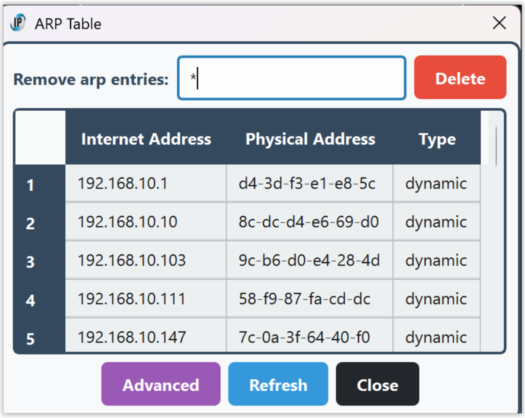
Tips:

The «Total» row always keeps counting, even if you reset the trip counter.

The «Trip Counter» row only counts while the timer is running. When it stops, the numbers turn red.

You can hover your mouse over any number to see more details.

Arp



The ARP Table dialog in IPGui lets you view and manage the Address Resolution Protocol (ARP) table on your Windows system.

What is the ARP Table?

The ARP table is a list that maps IP addresses to physical (MAC) addresses on your local network.

It is used by your computer to know which device (by MAC address) is associated with which IP address, allowing local network communication to work efficiently.

Features

View ARP Table:
See all current ARP entries in a clear table, showing:

- Internet Address: The IP address of a device on your local network.
- Physical Address: The MAC address of that device.
- Type: Whether the entry is dynamic (learned automatically) or static (manually set).

Remove ARP Entries:
You can delete a specific ARP entry by entering its IP address and clicking Delete, or delete all entries by entering *.

Note: Deleting entries may require administrator rights (UAC prompt).

Advanced/Basic View:
Toggle between a basic and advanced ARP table view.
The advanced view may show more technical details, depending on your system.

Refresh:
Instantly refresh the ARP table to see the latest entries.

Tooltips:
Hover over the IP input field for helpful instructions.

Why is this useful?

Troubleshooting:

If you have network issues, checking the ARP table can help you see if your computer is communicating with the correct devices.

Network Management:
Advanced users can clear or modify ARP entries to resolve conflicts or test network changes.

Security:
Spot unexpected devices on your network by their MAC and IP addresses.

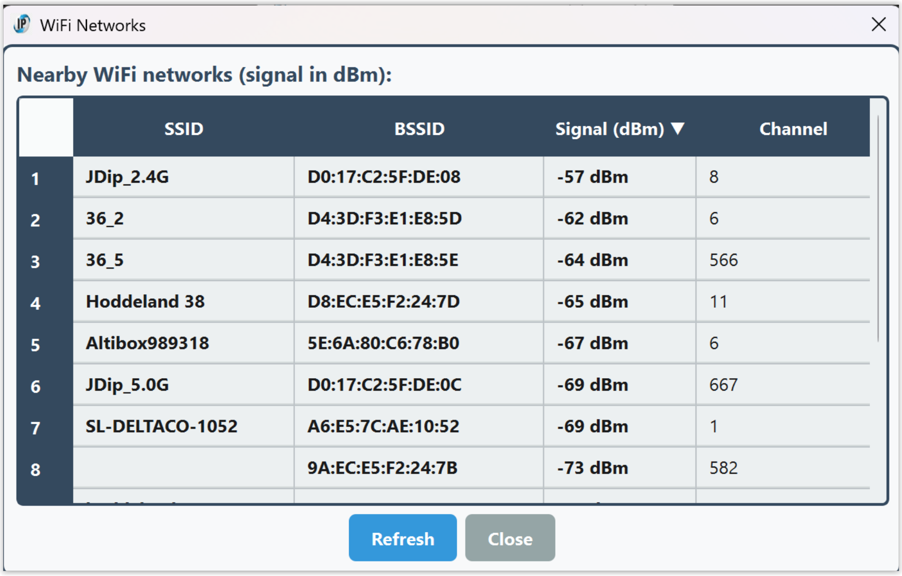
Tip:

If you’re unsure, just use the dialog to view your ARP table.

Deleting or modifying entries is for advanced users and usually not needed for everyday use.

Mouseover Help:
Hover your mouse over the IP input field to see a tooltip explaining how to delete a single entry or all entries.

WiFi scan



	SSID	BSSID	Signal (dBm) ▼	Channel
1	JDip_2.4G	D0:17:C2:5F:DE:08	-57 dBm	8
2	36_2	D4:3D:F3:E1:E8:5D	-62 dBm	6
3	36_5	D4:3D:F3:E1:E8:5E	-64 dBm	566
4	Hoddeland 38	D8:EC:E5:F2:24:7D	-65 dBm	11
5	Altibox989318	5E:6A:80:C6:78:B0	-67 dBm	6
6	JDip_5.0G	D0:17:C2:5F:DE:0C	-69 dBm	667
7	SL-DELTACO-1052	A6:E5:7C:AE:10:52	-69 dBm	1
8		9A:EC:E5:F2:24:7B	-73 dBm	582

The WiFi Scan dialog in **IPGui** lets you view all nearby wireless networks and their signal strengths in real time.

What does it do?

Scans for all WiFi networks your computer can see.

Shows each network's:

- SSID (network name)
- BSSID (unique hardware address of the access point)
- Signal strength (in dBm, higher is better)
- Channel (the radio channel used)

Features

Live Table:
The table updates automatically every few seconds, so you always see the latest list of available networks.

Sorting:
Click any column header (SSID, BSSID, Signal, Channel) to sort the list by that column. For example, click "Signal" to see the strongest networks at the top.

Manual Refresh:
Click Refresh to scan for networks immediately.

Tooltips:
Hover your mouse over a network name (SSID) or BSSID to see the full value, even if it’s too long to fit in the table.

No Configuration Needed:
This dialog is for viewing only; it does not connect, disconnect, or change your WiFi settings.

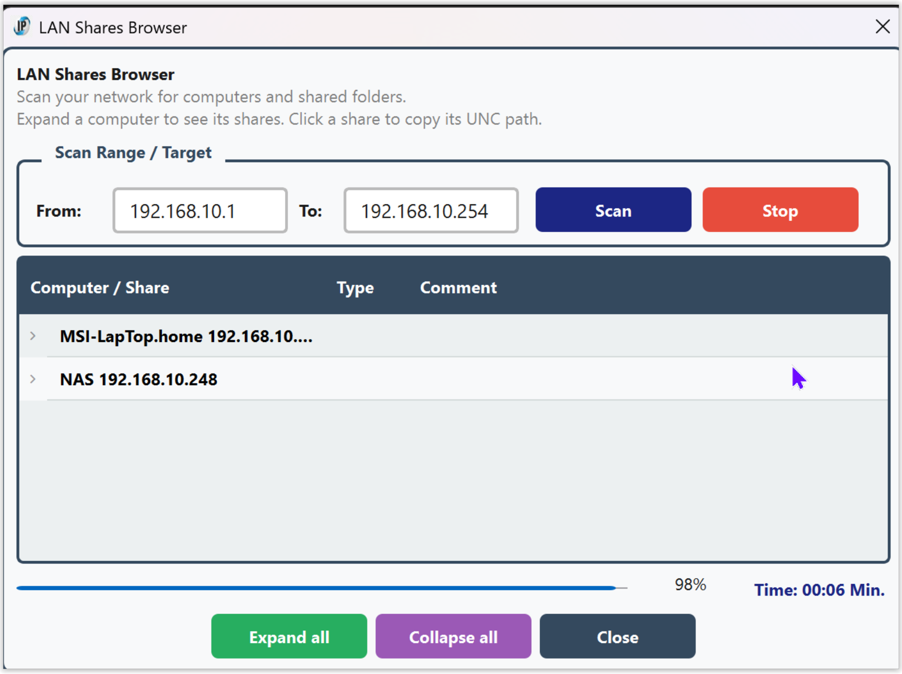
Why is this useful?

Find the best network:
Quickly see which WiFi networks have the strongest signal.

Troubleshoot connectivity:
See if your network is visible and how strong its signal is.

Network management:
Identify overlapping channels or nearby access points.

LAN Shares Browser



What it does:

The LAN Shares Browser scans your local network for computers and shows you any shared folders (also called «shares») that are available on those computers. You can easily see which computers are online, what their names are, and what folders they are sharing.

Why is it useful?

This tool helps you quickly find all shared folders on your network, so you don't have to guess which computers are sharing files or search manually. It's great for finding your NAS, shared PCs, or any device that offers Windows file sharing. You can easily copy the network path to any share and open it in File Explorer.

How to use it:

Open the tool:

Go to the menu and select Actions → NetTools → LAN Shares.

Set the scan range:

By default, the tool will scan your whole local network (for example, from 192.168.1.1 to 192.168.1.254).
If you want to scan a smaller range, just change the "From" and "To" IP addresses.

Start the scan:

Click the Scan button.
The tool will quickly check which computers are online, then look for shared folders on each one.
You'll see a progress bar and a timer showing how long the scan takes.

View results:
Computers with shares will appear in the list.
Click the arrow next to a computer to see its shared folders.
Shared folders are shown in blue for easy reading.

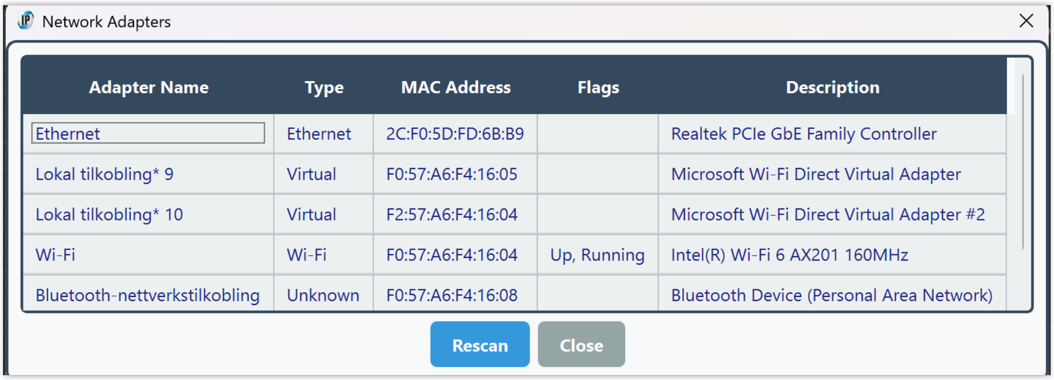
Copy a share path:
Double-click any shared folder to copy its network path (UNC path) to the clipboard.
You can then paste this path into File Explorer or another program to access the share.

Other features:
Use «Expand all» or «Collapse all» to open or close all computers in the list.
The «Close button» exits the tool. So does «[Ctrl]+W».

Tips:

The scan is very fast and will only show computers that are online and sharing folders.
If you don't see a computer you expect, make sure it's turned on and connected to the same network.
You can scan any range of IP addresses on your local network.

Network adapters



The Network Adapters dialog in **IPGui** gives you a complete overview of all network interfaces on your computer.

What does it show?

Adapter Name:
The friendly name of each network adapter (e.g., "Wi-Fi", "Ethernet", "VirtualBox Host-Only Network").

Type:
The kind of adapter (Wi-Fi, Ethernet, Virtual, PPP, or Unknown).

MAC Address:
The unique hardware address of the adapter.

Flags:
Status indicators such as Up, Running, Loopback, or P2P (Point-to-Point).

Description:
The full Windows description for the adapter, often including the manufacturer and model.

Features

Live Table:
All adapters are listed in a table with clear columns.
The table is automatically sized to fit all information, so you don't need to scroll horizontally.

Rescan:
Click Rescan to refresh the list and see any changes (for example, if you plug in a USB network adapter or connect/disconnect Wi-Fi).

Tooltips:
Hover over the description column to see the full adapter description, even if it's too long to fit in the table.

Color Coding:
All information is shown in a deep blue color for clarity and consistency.

No Editing:
The table is read-only, so you can't accidentally change any settings.

No Loopback:
Loopback adapters (used for internal networking) are hidden by default to keep the list relevant.

Why is this useful?

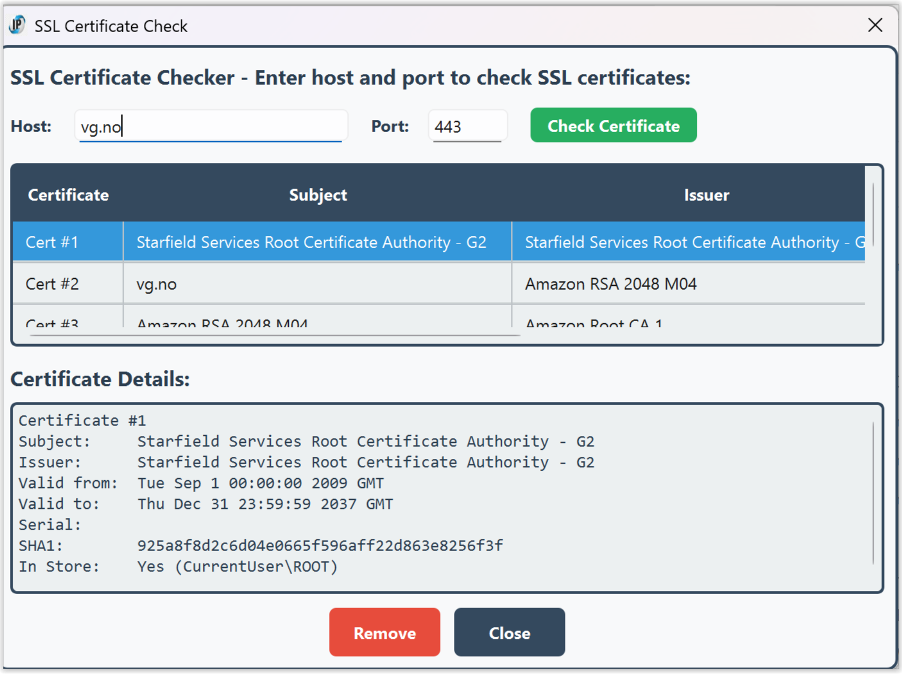
Troubleshooting:
Quickly see all network adapters, their status, and details—helpful for diagnosing connection problems.

Network Management:
Identify which adapter is which, especially if you have multiple network cards, USB Wi-Fi dongles, or virtual adapters.

Security:
Spot unexpected or virtual adapters that may have been added by software.

Tip:
If you add or remove a network adapter while the dialog is open, just click Rescan to update the list.
A popup will confirm when the list has been refreshed.

HTTPS Certificate Check



The HTTPS Certificate Check tool allows you to inspect the SSL/TLS certificates used by any HTTPS server. This is useful for verifying the security and validity of websites, troubleshooting certificate errors, or checking if a certificate is trusted by your Windows system.

Why is it useful?

Security Verification:
Ensure that the websites you visit use valid, trusted certificates and are not vulnerable to man-in-the-middle attacks.

Troubleshooting:
Diagnose problems with HTTPS connections, such as expired, self-signed, or untrusted certificates.

System Trust:
Check if a certificate is present in your Windows certificate store, which determines whether Windows and browsers trust the site.

Certificate Management:
Identify and, if needed, remove problematic or outdated certificates from your system (advanced users only).

How to Use

Open the dialog:
Select "HTTPS Certificate Check..." from the NetTools menu.

Enter the host and port:

Host:
Type the domain name or IP address of the server you want to check (e.g., www.google.com).

Port:
Enter the port number (default is 443 for HTTPS).

Start the check:
Click the Check button. The tool will connect to the server and retrieve its certificate chain.

View certificate details:
Each certificate in the chain is displayed with details such as Subject, Issuer, Validity period, Serial number, and SHA1 fingerprint.

The tool will indicate if a certificate is found in your Windows certificate store (Root or CA).

Remove a certificate (advanced):
If a certificate is found in your Windows store, a Remove button will appear.

Click Remove to delete the certificate from the store (requires administrator rights).

Warning! Removing trusted certificates may affect system security and connectivity.

Close the dialog:
Click Close or press Ctrl+W to exit the dialog.

Notes

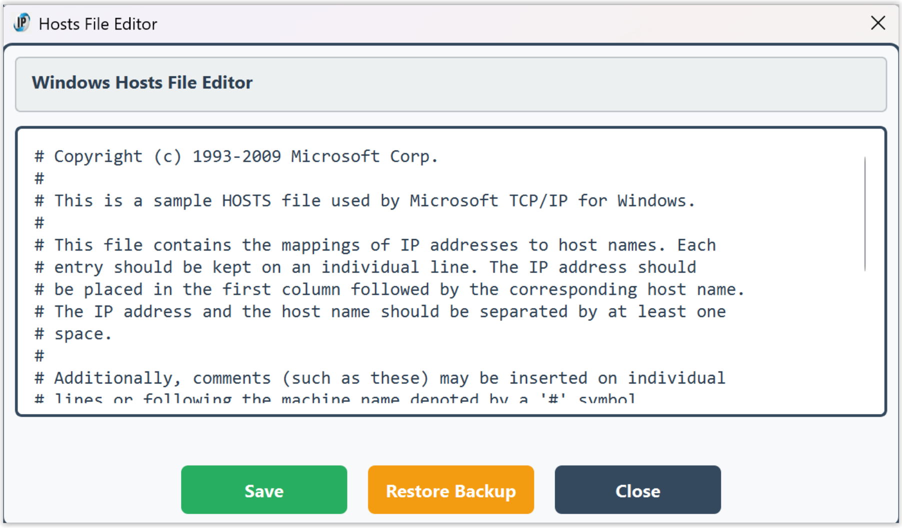
If the connection fails, an error message will be shown.

The certificate chain is displayed in order, with the server certificate first.

This tool is for inspection and troubleshooting only.

Do not remove certificates unless you are sure of what you are doing.

Hosts File Editor



What is the Hosts File?

The Windows hosts file is a system file that maps hostnames (like example.com) to IP addresses. It is used by Windows before DNS lookups and can be used to block sites, redirect domains, or test network changes.

Why is it Useful?

- Block unwanted sites: Redirect domains to 127.0.0.1 to block ads, malware, or distracting websites.
- Test website changes: Point a domain to a different IP for development or migration testing.
- Bypass DNS: Override DNS for specific domains, useful for troubleshooting or offline work.
- Quickly undo changes: The built-in backup/restore lets you safely experiment and revert.

How to Use

Open the Hosts File Editor:
Go to Actions → NetTools → Hosts File Editor.

View and Edit:
The editor displays the current contents of your hosts file. You can add, remove, or modify entries.

Each line should be in the format:

IP_address hostname [# comment]

Example:

127.0.0.1   localhost

192.168.1.10   myserver.local # test server

Save Changes:
Click «Save» to write your changes.

If administrator rights are needed, you will be prompted for elevation.

The app always creates a backup of the current hosts file before saving, so you can restore if needed.

Restore Backup:
Click «Restore Backup» to revert the hosts file to its previous state (before your last save).
This is useful if you want to undo recent changes or recover from mistakes.
No manual file handling is needed.

Close:
Click Close or press Ctrl+W to exit the editor.
If you have unsaved changes, you will be prompted to save or discard them.

Safety and Backup

Automatic Backup:
Every time you save, the app creates a backup of the hosts file as it was before your changes.
The backup is stored in:

C:\Users\Public\AppData\Local\IPGui\hosts.bak

Restore with One Click:
The Restore Backup button will restore the hosts file from this backup, even if administrator rights are needed.

No Risk of Permanent Loss:
If you make a mistake, you can always restore the previous version.

Notes

Administrator Rights:
Editing the hosts file usually requires administrator rights. The app will prompt for elevation if needed.

Antivirus/Antimalware:
Some security software may block changes to the hosts file. If saving fails, check your antivirus settings.

Portable Use:
The backup is stored in a public, non-user-specific folder, making it safe for portable use. There is a menu entry to clean up the temporary files in the main menu.

Troubleshooting

Cannot Save:
If you see a "Save Failed" message, try running the app as administrator or check for antivirus interference.

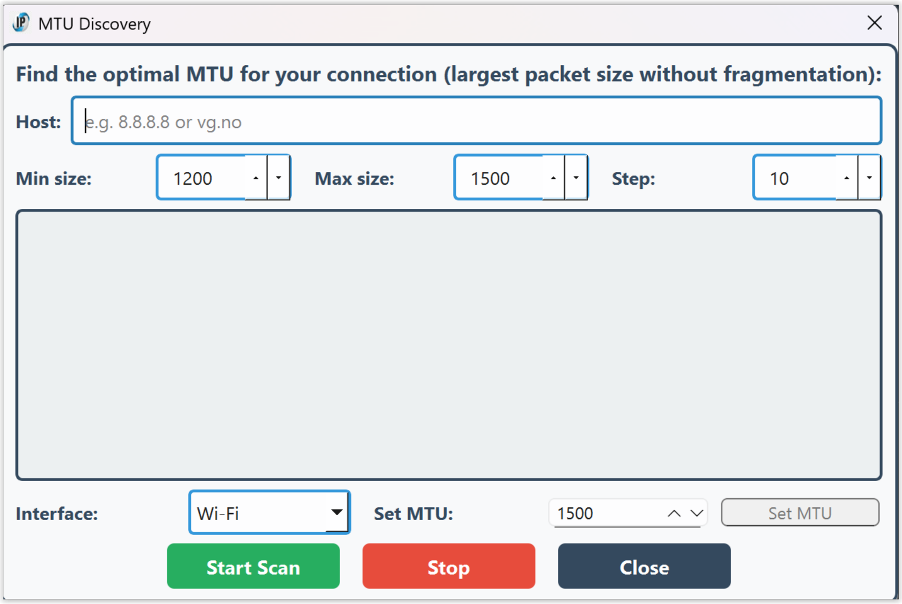
Backup Not Found:
If you try to restore but no backup exists, you will be notified.

Best Practices
Only add or change lines you understand.
Use the backup/restore feature to experiment safely.
Remember to save after making changes.

Tip:

You can use the Hosts File Editor to quickly block sites, test new servers, or undo accidental changes—without manually editing system files or using Notepad as administrator.

MTU Discovery



What does it do?

The MTU Discovery tool helps you find the largest packet size (MTU, or Maximum Transmission Unit) that can be sent to a remote host without fragmentation. It scans a range of packet sizes using the Windows ping command with the "Don't Fragment" flag, and reports which sizes succeed and which fail. You can then set your network adapter’s MTU to the optimal value directly from the tool.

Features

Automatic MTU Scan:
Test a range of packet sizes to find the largest value that works without fragmentation.

Customizable Range and Step:
Choose the minimum, maximum, and step size for the scan.

Detailed Results:
See which packet sizes succeed or fail, with color-coded output.

Interface Selection:
Pick which network adapter to set the MTU for.
Default selection is the active network card.

Editable MTU Setting:
After scanning, set the MTU to the discovered value or enter your own (with a warning if you exceed the discovered max).

One-Click MTU Change:
Applies the MTU setting using Windows tools (with UAC prompt if needed).

Fallback:
If automatic setting fails, opens the Windows network settings for manual adjustment.

Why is this useful?

Optimizes Network Performance:
Using the correct MTU can improve speed and reliability, especially for VPNs, gaming, or problematic connections.

Troubleshooting:
Helps diagnose issues caused by incorrect MTU, such as slow downloads, failed uploads, or web pages not loading.

Convenience:
No need to use command-line tools or search for the right adapter—everything is in one dialog.

How to use

- Open the MTU Discovery tool from the menu.
- Enter the host you want to test (default is 8.8.8.8, Google DNS).
- Set the range and step for packet sizes (e.g., 1200–1500, step 10).
- Click Start Scan.
- The tool will ping the host with increasing packet sizes and show which succeed.
- When the scan is done, the largest successful MTU is shown.
- Select your network interface from the dropdown.
- Edit the MTU value if you wish (the discovered max is pre-filled).
- If you enter a value above the discovered max, you'll get a warning.
- Click Set MTU to apply the new value.
- If successful, you'll see a confirmation.
- If not, you can open Windows network settings to set it manually.

Tips

Narrow Down:
Start with a large step (e.g., 100), then scan a smaller range with a smaller step (e.g., 10 or 1) for precision.

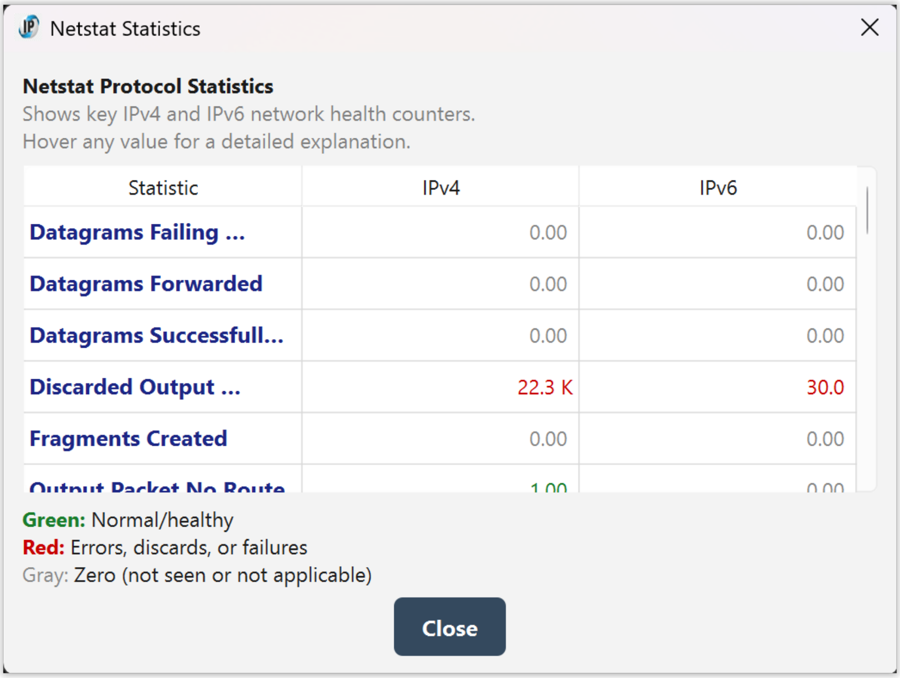
Test Different Hosts:
If you have issues with a specific site or service, try scanning to its server.

Reboot or Reconnect:
Some adapters may require a reconnect or reboot after changing MTU.

VPNs and Tunnels:
MTU may be lower when using VPNs or certain ISPs—scan again if your network changes.

Admin Rights:
Setting MTU requires administrator privileges; accept the UAC prompt when asked.

Netstat Statistics



What is Netstat Statistics?

The Netstat Statistics tool displays detailed counters about your computer's network activity, as reported by the Windows netstat -s command. It shows key statistics for both IPv4 and IPv6 protocols, including packets sent and received, errors, discards, and fragmentation events. These counters provide a snapshot of your system's network health and activity.

Typical Uses

- Diagnosing network problems: Identify if your system is experiencing packet loss, errors, or routing issues.
- Monitoring network health: Quickly see if your network is operating normally or if there are abnormal error rates.
- Troubleshooting connectivity: Determine if issues are due to local configuration, hardware, or external factors.
- Comparing IPv4 and IPv6 activity: See which protocol is being used more and if either is experiencing problems.

How to Use

Open the Netstat Statistics dialog:
Go to Actions → NetTools → Netstat Statistics.

View the statistics table:
The dialog displays a table with all key IPv4 and IPv6 counters.
Numbers are humanized (e.g., 1.2K, 3.4M) for readability.
Each statistic has a tooltip: hover over any value or name for a detailed explanation.

Interpret the colors:

Green: Normal/healthy values (e.g., packets delivered).

Red: Errors, discards, or failures (pay attention to these).

Gray: Zero (not seen or not applicable).

Close the dialog:
Click Close or press Ctrl+W.

Why is it Useful?

Q**uickly spot network issues:**
High error, discard, or failure counts may indicate hardware problems, bad cables, driver issues, or misconfiguration.

Monitor network performance:
See if your system is forwarding packets (acting as a router), or if it's dropping packets due to congestion.

IPv4/IPv6 comparison:
Helps you understand which protocol is active and if either is experiencing issues.

Safe and read-only:
Viewing statistics does not affect your network or require administrator rights.

Best Practices & Tips

Check regularly:
Use this tool periodically to monitor your network's health, especially after changes or when troubleshooting.

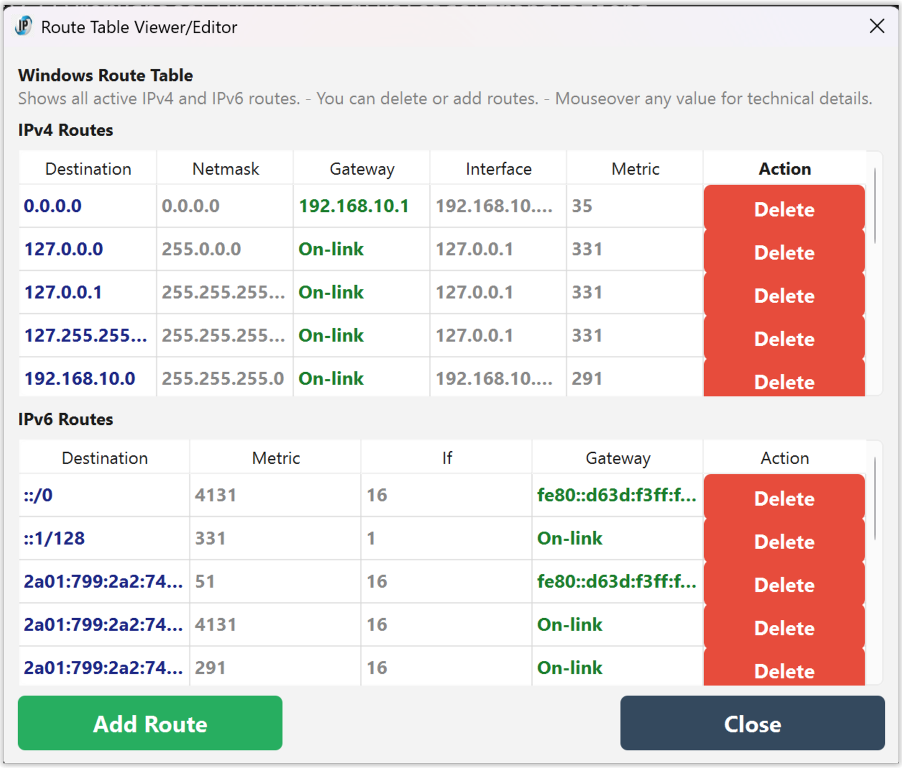
Look for non-zero errors:
Any non-zero value in red fields (errors, discards, failures) may warrant further investigation.

Reset counters:
To see the effect of recent changes, you can reboot your system or disable/enable your network adapter to reset these counters.

Hover for help:
Every statistic has a tooltip with a clear explanation—hover over any value or name for details.

Compare IPv4 and IPv6:
If you see issues in one protocol but not the other, it may help narrow down the source of a problem.

Route Table Viewer/Editor



What is the Route Table Viewer/Editor?

The Route Table Viewer/Editor displays the current Windows routing tables for both IPv4 and IPv6.

It shows all active routes used by your computer to determine how network traffic is sent and received.

You can view detailed information for each route, including destination, gateway, interface, and metric.

Typical Uses

- **Diagnosing network problems:**
  See how your computer decides where to send packets, and spot misconfigurations.
- **Checking VPN or multi-homing setups:**
  Verify which routes are active when using VPNs or multiple network adapters.
- **Consulting for advanced troubleshooting:**
  Network consultants may need to add, remove, or adjust routes for complex setups.

How to Use

Open the Route Table Viewer/Editor:
Go to Actions → NetTools → Route Table Viewer/Editor.

View the tables:
The dialog shows two tables: one for IPv4 routes, one for IPv6 routes.

Each table lists all active routes with columns for destination, gateway, interface, and metric.

Mouseover any value for a technical explanation (geek info).

Colors:

Blue: Destination addresses.

Green: Gateway addresses.

Gray: Other fields.

Delete or add routes (advanced):

Click «Delete» to remove a route (requires administrator rights).

Use the «Add Route» section to create a new route (requires administrator rights).

Warning: Editing routes is for advanced users only.

Mouseover the buttons for safety tips.

Close the dialog:

Click Close or press «[Ctrl]+W».

Why is it Useful?

Network troubleshooting:
Quickly see if your traffic is routed as expected, or if there are unwanted or missing routes.

VPN and remote access:
Check which routes are added or removed when connecting to a VPN.

Advanced configuration:
Allows consultants and admins to fine-tune routing for special requirements.

Best Practices & Tips

Viewer mode is safest:
Most users should use this tool only to view routes, not to edit them.

Editing routes:
Only add or delete routes if you know exactly what you are doing.
Incorrect changes can break network connectivity.

Mouseover for help:
Every column and button has a tooltip with technical details and safety tips.

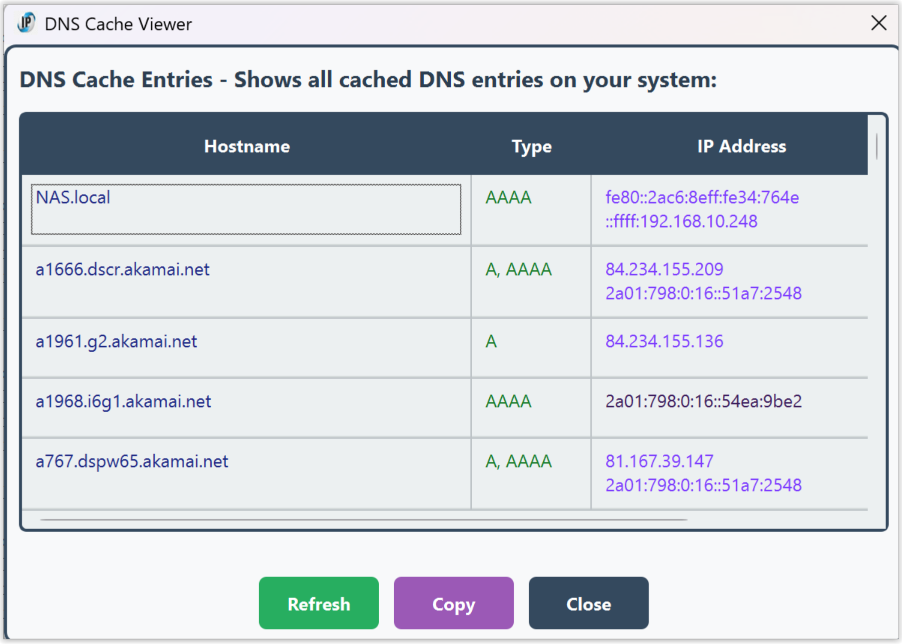
Refresh if in doubt:
If you suspect the table is out of date, close and reopen the dialog to reload the latest routes.

System routes:
Some routes (especially in IPv6) may be automatically regenerated by Windows.
Deleting them may not be permanent.

Tip:

If you are not a network specialist, use this tool as a viewer only.
For editing, consult your IT department or a network professional.

DNS Cache Viewer



Menu:
Actions → NetTools → DNS Cache Viewer

What it does

The DNS Cache Viewer displays all DNS records currently cached by Windows on your system. This includes hostnames, record types (A, AAAA, CNAME, etc.), resolved IP addresses, time-to-live (TTL) values, and the status of each entry (such as Success or Negative).

The table is color-coded for clarity:

Blue: Hostname
Green: Record type and valid status
Purple: IPv4 addresses
Dark purple: IPv6 addresses
Gray: "Not found" markers
Red: Negative/failed entries

You can sort the table by clicking any column header. Mouse over any value for a technical explanation.

Why is it useful?

DNS cache entries show which hostnames your system has recently resolved and what IP addresses they mapped to. This is invaluable for:

Troubleshooting:
See if a domain is cached incorrectly or if negative responses are causing connectivity issues.

Diagnostics:
Quickly check if a hostname is resolving as expected without using command-line tools.

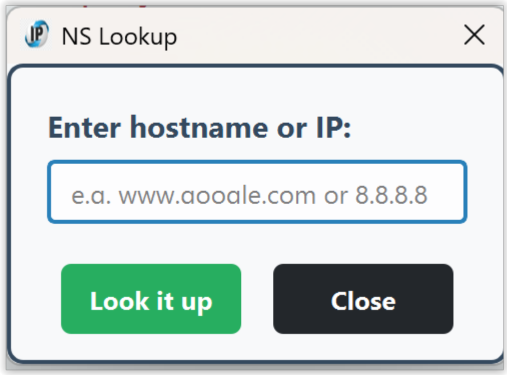
Privacy/Security:
Spot unexpected or suspicious domains in your cache.

Tip

If you don’t see expected entries, try browsing some websites or running NS Lookup in this app. Then reopen the viewer.

You can flush the DNS cache from the main window to clear all entries and start fresh.

NS Lookup



Menu: Actions → NetTools → NS Lookup

What it does

The NS Lookup tool allows you to query DNS records for any hostname or IP address. Enter a domain name (like www.example.com) or an IP address, and the tool will display the DNS server used, the resolved canonical name, and all associated IPv4 and IPv6 addresses.

The results are shown in a clear, monospace format for easy reading and copying.

Why is it useful?

Troubleshooting:
Quickly check if a domain resolves correctly from your current network.

Diagnostics:
See which DNS server is being used and what addresses are returned.

Verification:
Confirm that DNS changes (such as new records or updates) have propagated.

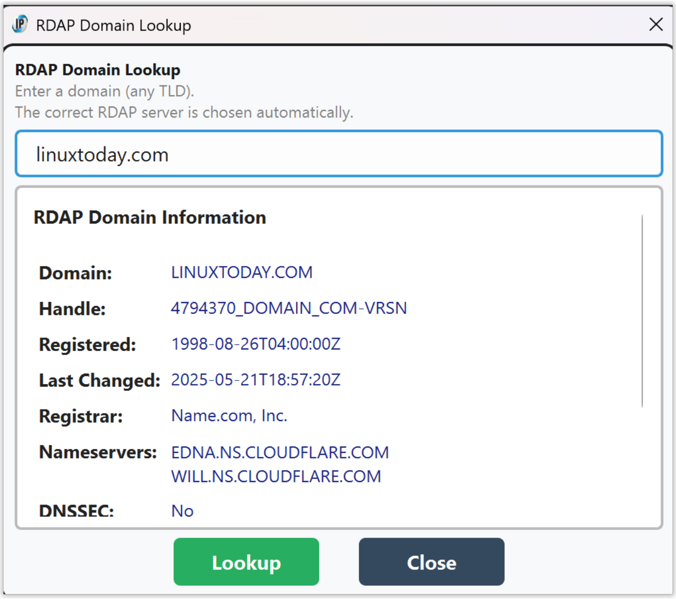
Learning:
Understand how DNS resolution works and what information is returned for different queries.

Tip

If you get no results or unexpected answers, try flushing your DNS cache (Actions → Flush DNS) and running the lookup again.

You can also use this tool to check reverse lookups by entering an IP address.

Whois Lookup



What it does:

The Whois Lookup tool lets you look up information about any domain name, such as who owns it, when it was registered, and which company manages it. This is useful if you want to find out who is behind a website or check when a domain will expire.

How to use it:

Open the main window and click the menu:

Actions → NetTools → Whois Lookup

In the dialog that appears, enter the domain name you want to look up (for example: example.com or www.google.com).

Click the «Lookup» button.
Wait a few seconds while the tool contacts the correct Whois server for the domain.

The results will appear in a table, showing details like:

- Domain name
- Registrar (the company that manages the domain)
- Registration and last changed dates
- Contact emails and phone numbers (clickable)
- Nameservers
- DNSSEC status
- Notices and legal information (with clickable links)

What is the difference between Whois Lookup and RDAP Lookup?

When you use the Whois Lookup feature in **IPGui**, you are actually using the modern RDAP (Registration Data Access Protocol) system behind the scenes. RDAP is the new, standardized way to look up domain registration information, replacing the older "Whois" system.

Whois is the traditional method for checking who owns a domain name, but it can be inconsistent and sometimes hard to read.

RDAP is the new standard, designed to provide the same information in a clearer, more secure, and more reliable way. It is supported by all major domain registries.

In IPGui, the Whois Lookup tool automatically uses RDAP for all supported domains.

This means you get up-to-date, accurate information, and you don't have to worry about which system is being used—the tool handles it for you.

In summary:

"Whois Lookup" in **IPGui** = "RDAP Lookup" (modern, automatic, and reliable)

You get the best available data for any domain, with clickable links and easy-to-read results.

Tips:

You can click on any email address or website link in the results to open them directly.

If the tool cannot find information for a domain, check that you typed it correctly.

This tool works for all domain endings (TLDs), including .com, .net, .org, .no, .co.uk, and more.

When to use it:

- To check who owns a website or domain.
- To see when a domain was registered or when it will expire.
- To find contact information for a domain owner or registrar.

Troubleshooting:

If you see a message about not being able to download the TLD list, make sure you are connected to the internet. This only happens if the list has not been downloaded, so it can mostly only happen at first use on said PC.

Some domains may have privacy protection, so not all contact details will always be visible.